



MAARULA CLASSES

TARGET- NIMCET / CUET.PG By: Amit Katiyar (MCA-JNU)

PERMUTATION & COMBINATION

NIMCET PYQ

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1. T_n represent the no. of triangles formed from a polygon having n vertices. $T_{n+1} - T_n = 21$, then n is equal to:
NIMCET 2007
(a)6 (b)7 (c)8 (d)none
2. How many different paths in the xy -plane are there from $(1, 3)$ to $(5, 6)$ if a path proceeds one step at a time by going either one step to the right (R) or one step upwards (U)?
NIMCET 2009
(a)35 (b)40
(c)45 (d)None of these
3. There are 10 points in a plane. Out of these, 6 are collinear. The number of triangles formed by joining these points is
NIMCET 2009
(a)100 (b)120
(c)150 (d)None of these
4. A set contains $(2n + 1)$ elements. If the number of subjects which contains atmost n elements is 4096, then the value of n is
NIMCET 2009
(a)28 (b)21 (c)15 (d)6
5. A polygon has 44 diagonals, then the number of its sides is
NIMCET 2011
(a)9 (b)10 (c)11 (d)12
6. The number of ways of forming different nine digit numbers from the number 223355888 by rearranging its digit, so that the odd digits occupy even positions is
NIMCET 2011
(a)16 (b)36 (c)60 (d)180
7. The number of words that can be formed by using the letters of the word 'MATHEMATICS' that start as well as end with T is
NIMCET 2012
(a)80720 (b)90720 (c)20860 (d)37528
8. The number of different license plates that can be formed in the format 3 English letters (A...Z) followed by 4 digits (0, 1.....9) with repetition allowed in letters and digit is equal to
NIMCET 2012
(a) $26^3 \times 10^4$ (b) $26^3 + 10^4$
(c)36 (d) 26^3
9. In which of the following regular polygons, the number of diagonals is equal to number of sides?
NIMCET 2012
(a)Pentagon (b)Square
(c)Octagon (d)Hexagon
10. If T_n , denotes the number of triangles which can be formed by using the vertices of a regular polygon of n sides. If $T_{n+1} - T_n = 21$, then n equals
NIMCET 2013
(a)5 (b)7 (c)6 (d)4
11. In how many different ways can the letters of the word 'CORPORATION' be arranged, so that all the vowels always come together?
NIMCET 2013
(a)810 (b)1440 (c)2880 (d)50400
12. The number of non-negative integers less than 1000 that contain the digit 1 are
NIMCET 2013
(a) 9^2 (b) 9^3 (c) $10^2 - 9^2$ (d) $10^3 - 9^3$
13. The total number of numbers that can be formed using the digits 3, 5 and 7 only, if no repetitions are allowed, is
NIMCET 2013
(a)39 (b)105 (c)15 (d)27
14. A password consists of two alphabets from English followed by three digits chosen from 0 to 3. Repetitions are allowed. The number of different password is
NIMCET 2014
(a) $^{25}P_1 \cdot ^{25}P_2 \cdot ^4P_1 \cdot ^2P_1$
(b) $(^{26}P_1)^2 \cdot (^4P_1)^3$
(c) $^{26}P_1 \cdot ^{26}P_2 \cdot ^4P_1 \cdot ^4P_2 \cdot ^4P_3$
(d) $(^{26}P_1 \cdot ^4P_1)$
15. The number of ways in which 5 days can be chosen each of the 12 months of a non-leap year, is
NIMCET 2014
(a) $(^{30}C_5)^4 \cdot (^{31}C_5)^7 \cdot (^{28}C_5)$
(b) $(^{30}C_5)^6 \cdot (^{28}C_5)^6$
(c) $(^{30}C_5)^7 \cdot (^{31}C_5)^4 \cdot (^{28}C_5)$
(d) $(^{30}C_5)^5 \cdot (^{31}C_5)^6 \cdot (^{28}C_5)$
16. The number of ways to arrange the letters of the English alphabet, so that there are exactly 5 letters between a and b, is
NIMCET 2014
(a) $^{24}P_5$ (b) $^{24}P_5 \cdot 20!$
(c) $^{24}P_5 \cdot 20! \cdot 2$ (d) $^{24}P_5 \cdot 24! \cdot 2$
17. There are 8 students appearing in an examination of which 3 have to appear in Mathematics paper and the remaining 5 in different subjects. Then, the number of ways they can be made to sit in a row, if the candidate in the Mathematics cannot sit next to each other is
NIMCET 2014
(a)2400 (b)16200 (c)4200 (d)14400
18. How many even integers between 4000 and 7000 have four different digits?
NIMCET 2014
(a)672 (b)840 (c)504 (d)728
19. If $42(^nP_2) = ^nP_4$, then the value of n is
NIMCET 2015
(a)2 (b)4 (c)9 (d)42
20. The number of bit strings of length 10 that contains either five consecutive 0's or five consecutive 1's is
NIMCET 2015
(a)64 (b)112 (c)220 (d)222





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21. There are n equally spaced points $1, 2, \dots, n$ marked on the circumference of a circle. If the point 15 is directly opposite of the point 49, then the total number of points is **NIMCET 2016**
(a)50 (b)68 (c)66 (d)70
22. The number of 5 people groups that can be selected from 9 people when two particular persons are not to be in the same group is **NIMCET 2016**
(a)126 (b)35 (c)91 (d)252
23. There are 4 books on fairy tales, 5 novels and 3 plays. In how many ways can they be arranged in the order, books on fairy tales, novels and they plays so that the books of the same category are put together? **NIMCET 2016**
(a)17280 (b)103680 (c)51840 (d)360
24. The permutations of (a, b, c, d, e, f, g) are listed in lexicographic order. Which of the following permutations are just before and just after the permutation bacdegf? **NIMCET 2016**
(a)agfedbc and bacdfge
(b)agfedcb and badcefg
(c)agfebcd and bacedgf
(d)agfedcb and bacdegf
25. m distinct animals of a circus have to be placed in m cages, one in each cage. If n ($n < m$) cages are too small to accommodate p ($n < p < m$) animals, then the number of ways of putting the animals in to cages are **NIMCET 2017**
(a) $\{(m-n)P_p\} \{(m-p)P_{m-p}\}$
(b) $(m-n)C_p$
(c) $\{(m-n)C_p\} \{(m-p)C_{m-p}\}$
(d) $(m-n)P_p$
26. 9 balls are to be placed in 9 boxes and 5 of the balls cannot fit into 3 small boxes. The number of ways of arranging one ball in each of the boxes is **NIMCET 2018**
(a)18720 (b)18270 (c)17280 (d)12780
27. A student council has 10 members. From this one President, one Vice-President, one Secretary, one Joint-secretary and two Executive Committee members have to be elected. In how many ways this can be done? **NIMCET 2018**
(a)151200 (b)75600 (c)37800 (d)18900
28. The number of natural members which are smaller than 2×10^8 and which contain only the digits 1 and 2 is **NIMCET 2018**
(a)786 (b)666 (c)766 (d)1066
29. If all the words, with or without meaning, are written using the letters of the word QUEEN and are arranged as in English. Dictionary, then the position of the word QUEEN is **NIMCET 2019**
(a)47th (b)44th (c)45th (d)46th
30. How many words can be formed starting with letter D talking all letters from word DELHI so that the letters are not repeated. **NIMCET 2020**
(a)4 (b)12 (c)24 (d)120
31. Naresh has 10 friends, and he wants to invite 6 of them to a party. How many times will 3 particular friends never attend the party? **NIMCET 2020**
(a)8 (b)7 (c)720 (d)35
32. There is a young boy's birthday party in which 3 friends have attended. The mother has arranged 10 games where a prize is awarded for a winning game. The prizes are identical. If each of the 4 children receives at least one prize, then how many distributions of prizes are possible? **NIMCET 2020**
(a)80 (b)84 (c)70 (d)72
33. There are 50 questions in a paper. Find the number of ways in which a student can attempt one or more questions. **NIMCET 2021**
(a) $2^{50} - 1$ (b) $2^{50} - 2$
(c) $2^{50} + 2$ (d) $2^{50} + 1$
34. A polygon has 44 diagonals, the number of sides are **NIMCET 2021**
(a)11 (b)10 (c)9 (d)12
35. In an examination of nine papers, a candidate has to pass in more papers than the number of papers in which he fails in order to be successful. The number of ways in which he can be unsuccessful is **NIMCET 2023**
(a)255 (b)256 (c)128 (d) $9 \times 8!$
36. Lines $L_1, L_2, \dots, L_{10}$ are distinct among which the lines $L_2, L_4, L_6, L_8, L_{10}$ are parallel to each other and the lines L_1, L_3, L_5, L_7, L_9 pass through a given point C. The number of point of intersection of pairs of lines from the complete set $L_1, L_2, L_3, \dots, L_{10}$ is: (P & C) **NIMCET 2024**
(a) 24 (b) 25
(c) 26 (d) 27
37. There are 9 bottles labelled $1, 2, 3, \dots, 9$ and 9 boxes labelled $1, 2, 3, \dots, 9$. The number of ways one can put these bottles in the boxes so that each box gets one bottle and exactly 5 bottles go in their corresponding numbered boxes is (P & C) **NIMCET 2024**
(a) $9 \times {}^9C_5$ (b) $5 \times {}^9C_5$
(c) $25 \times {}^9C_5$ (d) $4 \times {}^9C_5$





38. A committee of 5 is to be chosen from a group of 9 people. The probability that a certain married couple will either serve together or not at all is (P & C)

- (a) $\frac{5}{9}$ (b) $\frac{1}{2}$
(c) $\frac{2}{3}$ (d) $\frac{4}{9}$

NIMCET 2024

39. Number of three digit numbers that can be formed using 0, 1, 2, 3 and 5 where these digits are allowed to repeat any number of times, is equal to

- (a) 125 (b) 100
(c) 115 (d) 140

NIMCET 2025

40. Number of permutations of the letters of the word BANGLORE such that the string ANGLE appears together in all permutations, is

- (a) 720 (b) 24
(c) 1228 (d) 120

NIMCET 2025

41. The number of 3-digit integers that are multiple of 6 which can be formed by using the digits 1,2,3,4,5,6 without repetition is

- (a) 22 (b) 26
(c) 20 (d) 24

NIMCET 2025

ANSWER KEY

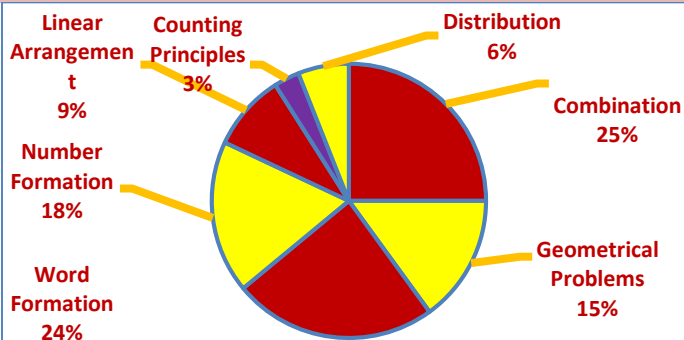
1.	a	2.	a	3.	a	4.	d	5.	c
6.	c	7.	b	8.	a	9.	a	10.	b
11.	d	12.	d	13.	c	14.	b	15.	a
16.	c	17.	d	18.	d	19.	c	20.	d
21.	b	22.	c	23.	b	24.	d	25.	a
26.	c	27.	b	28.	c	29.	d	30.	c
31.	b	32.	b	33.	a	34.	a	35.	b
36.	c	37.	a	38.	d	39.	b	40.	b
41.	d								

Analysis 2007 to 2025

Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	01	NIMCET 2016	04
NIMCET 2008	00	NIMCET 2017	01
NIMCET 2009	03	NIMCET 2018	03
NIMCET 2010	00	NIMCET 2019	01
NIMCET 2011	02	NIMCET 2020	03
NIMCET 2012	03	NIMCET 2021	02
NIMCET 2013	04	NIMCET 2022	00
NIMCET 2014	05	NIMCET 2023	01
NIMCET 2015	02	NIMCET 2024	03
		NIMCET 2025	03

SUB TOPICWISE ANALYSIS



Sub Topic wise Questions Asked in NIMCET

Sub Topic	Qs. Asked
Combination	09
Geometrical Problems	05
Word Formation	08
Number Formation	06
Linear Arrangement	03
Counting Principles	01
Distribution	02

SOLUTION

1. (a)

As per the given condition.

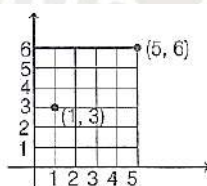
T_n triangle can be formed by using the vertices of a regular polygon of n .

$$\therefore {}^{n+1}C_3 - {}^nC_3 = 21$$

$$\Rightarrow \frac{(n+1)(n)(n-1)}{1 \cdot 2 \cdot 3} - \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} = 21 \Rightarrow \frac{n(n-1)}{2} = 21$$

Clearly, $n = 7$

2. (a)



To go from (1,3) to (5,6) total of 7 steps are needed out of which any 4 will be in right and remaining 3 in upward, total 7 steps in which 4R,3U (here seven things in which 4 right steps are identical, 3 upward steps are identical) hence total no. of ways

$$= \frac{7!}{4!3!} = 35$$

3. (a) Number of triangles = (Number of ways of selecting 3 points from given 10 points) - (Number of ways selecting 3 points from 6 collinear points).

$${}^{10}C_3 - {}^6C_3 = 120 - 20 = 100$$

4. (d) The number of required subsets are

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + \dots + {}^{2n+1}C_n$$

We should know

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + \dots + {}^{2n+1}C_n$$

$$= \frac{1}{2} [2^{2n+1}] = 2^{2n} = 4^n$$

But

$$4^n = 4096 = 4^6 \Rightarrow n = 6$$



5. (c) Let a polygon has n side

$$\Rightarrow {}^nC_2 - n = 44$$

$$\Rightarrow \frac{n(n-1)}{2} - n = 44$$

$$\Rightarrow n(n-3) = 88$$

$$\Rightarrow n(n-3) = 11(11-3) \Rightarrow n = 11$$

6. (c) Given number is 223355888.

Here, digits are (2,2,3,3,5,5,8,8,8)

Here, odd digits are (3,3,5,5) and even digits are (2,2,8,8,8)

In a nine digit number, there are four even position in which four odd number (3,3,5,5) can be arranged in $\frac{4!}{2! \times 2!} = 6$ and rest of five odd position, five even number (2,2,8,8,8) can be arranged in $\frac{5!}{2! \times 3!} = 10$

$$\therefore \text{Total number of ways} = 6 \times 10 = 60$$

7. (b)

T									T
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Total number of letters in the word "MATHEMATICS" is 11 and there are 2 T's, 2 A's and 2 M's

Since T is fixed at starting and at the ending,
 $\therefore$ Required number of ways

$$= \frac{9!}{2!2!} = \frac{362880}{2 \cdot 2} = 90720$$

8. (a) The number of arrangements of 3 English letters with repetitions allowed $= 26 \cdot 26 \cdot 26 = (26)^3$

The number of arrangements of 4 digits with repetition allowed $= 10 \cdot 10 \cdot 10 \cdot 10 = (10)^4$

$\therefore$ Required number of different licence plates $= (26)^3 \times (10)^4$ (According to given option)

9. (a) Let number of sides in the polygon be n then number of its diagonal will be $\frac{n(n-3)}{2}$

But it is given that, $n = \frac{n(n-3)}{2}$

$$\Rightarrow 2n = n(n-3)$$

$$\Rightarrow 2 = n-3 \Rightarrow n = 5$$

Hence polygon will be pentagon.

10. (b) As per the given condition.

T_n triangle can be formed by using the vertices of a regular polygon of n .

$$\therefore {}^{n+1}C_3 - {}^nC_3 = 21$$

$$\Rightarrow \frac{(n+1)(n)(n-1)}{1 \cdot 2 \cdot 3} - \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} = 21 \Rightarrow \frac{n(n-1)}{2} = 21$$

Clearly, $n = 7$

11. (d) In word 'CORPORATION', we treat the vowels OOAIO as one letter.

This, we have CRPRTN (OOAIO).

This has $7(6+1)$ letters of which R occurs 2 times and the rest are different.

Number of ways arranging these letters

$$= \frac{7!}{2!} = 2520$$

Now, 5 vowels in which O occurs 3 times and the rest are different, can be arranged in $\frac{5!}{3!} = 20$ ways.

$$\text{Required number of ways} = (2520 \times 20) = 50400$$

12. (d) Total numbers which contains digit 1 = total number of numbers - total numbers which does not contains digit 1 $= 10^3 - 9^3$
 So, the numbers containing 1 is $10^3 - 9^3$.

13. (c) Single digit number = 3

Two digit numbers are $3 \times 2 = 6$

Three digit numbers are $3 \times 2 \times 1 = 6$

Hence, total numbers are 15.

14. (b) Total alphabet in English = 26

Two words can be selected from 26 alphabet in 26×26 ways.

Now, next three digits from 0 to 3

i.e. 0,1,2,3 are selected.

$$4 \times 4 \times 4 = (4)^3$$

$$\text{The number of password} = 26^2 \times 4^3 = ({}^{26}P_1)^2 \times ({}^4P_1)^3$$

15. (a) In a leap year, 7 months have 31 days

4 months have 30 days and 1 month (Feb) 28 days

$$\text{Total number of ways} = ({}^{30}C_5)^4 \cdot ({}^{31}C_5)^7 \cdot ({}^{28}C_5)$$

16. (c) There are 26 letters in the English alphabet. If we separate the group (a, some 5 letters, b), we will be left with 19 more letters. These 20 objects (1 group + 19 letters) can be arranged among themselves in $20!$ ways.

Since either a or b can be at the beginning or the end of the group of 7 letters (a, some 5 letters, b), the number of possible arrangements of the group will be $= 2 \times 5!$

Also, each group of 5 letters can be selected from the remaining 24 letters (except a and b) in ${}^{24}C_5$ ways.

Required total number of ways

$$= (2 \times 5! \times {}^{24}C_5) \times 20!$$

$$= 2 \times {}^{24}P_5 \times 20!$$

17. (d) Total number of students = 8

5 different subjects candidates can be seated in ${}^5P_5 = 5!$ ways

In between 5 candidates there are six places for 3 Mathematics candidates

$\therefore$ The Mathematics candidates can be seated in 6P_3 ways

$\therefore$ By fundamental principle of counting.

The required number of ways

$$= 5! \times {}^6P_3 = 120 \times \frac{6!}{3!}$$

$$= 120 \times 6 \times 5 \times 4 = 14400$$

18. (d) Between 4000 and 5000, choose unit digit in 4 ways $(0,2,6,8) = 4$

Choose tenth place of remaining 8 numbers = 8 (because 2 even number out of 10 numbers are already taken).

[As number start from 4 and end with even digit].

Choose hundred place by remaining 7 numbers = 7 ways.

Total number ways between 4000 to 5000

$$= 4 \times 8 \times 7 = 224.$$

Similarly, for 5000 to 6000.

Unit digit is chosen in 5 ways $(0,2,6,8,4) = 5$

Tenth digit is chosen in 8 ways = 8



Thousandth digit is chosen by remaining 7 numbers
= 7 ways.

Total number ways = $5 \times 8 \times 7 = 280$

Similarly, for 6000 to 7000 .

Total number ways $4 \times 8 \times 7 = 224$

Hence, total number ways for 4000 to 7000
= $224 + 280 + 224 = 728$.

19. (c)

We have, $42 {}^n P_2 = {}^n P_4$

$$\Rightarrow 42 \frac{n!}{(n-2)!} = \frac{n!}{(n-4)!} \Rightarrow \frac{(n-2)!}{(n-4)!} = 42$$

$$\Rightarrow \frac{(n-2)(n-3)(n-4)!}{(n-4)!} = 42 \Rightarrow n^2 - 5n + 6 = 42$$

$$\Rightarrow n^2 - 5n = 36 \Rightarrow (n-9)(n+4) = 0 \Rightarrow n = 9$$

20. (d) Observe first 5 consecutive 0 's

Summation rule : the first 5 consecutive 0 's could start at position 1,2,3,4,5, or 6 .

Start at the first position-

Other 5 bits can be anything $2^5 = 32$

Start at the second position-

The first bit must be a 1 .

There are various possibilities that can be included in it.

Remaining bits can be anything $2^4 = 16$

Start at third position.

The second bit must be a 1 due to above-mentioned reason.

First bit and last 3 bits can be anything $2^4 = 16$

Starting at 4,5 and 6 positions.

Same as starting at positions 2 or 3, 16 each

Total = $32 + 16 + 16 + 16 + 16 + 16 = 112$

The five consecutive 1's ensue the same pattern and have different like 112 possibilities.

There would be two cases counted twice (that we thus need to excluded) 0000011111 and 1111100000 .

Total = $112 + 112 - 2 = 222$

21 (b) Directly opposite points are 49 and 15

Difference of directly opposite point = $49 - 15 = 34$

Number of points on right side = $34 - 1 = 33$

Number of points on left side = $34 - 1 = 33$

Total number of points = $33 + 33 + 2 = 68$

22 (c) Total number of peoples = 9

Number of ways in which 5 people groups can be selected from 9 people. According to the question, (5 people are selected from 9)-(2 particulars people already in the same group).

$$\Rightarrow {}^9 C_5 - {}^7 C_3 \Rightarrow \frac{9!}{5!4!} - \frac{7!}{3!4!} \Rightarrow \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2} - \frac{7 \times 6 \times 5}{3 \times 2}$$

$$\Rightarrow 126 - 35 \Rightarrow 91$$

23 (b) 4 books arranged = 4!

5 Novels arranged = 5!

3 plays arranged = 3!

all of them together arranged

$$3!(4! \times 5! \times 3!) = 103680$$

24 (d) Lexicographic order is the order in which the words appear in the dictionary.

For the word before and after check starts from the left end.

Set is $\{a, b, c, d, e, f, g\}$

The word before bacdefg is agfedcb.

The word after bacdefg is bacdegf.

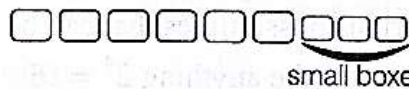
From the option, of all the given options, (a), (b), (c) are different from given order of word.

25. (a) Number of ways of arranging p big animals into $m - n$ big cages = ${}^{m-n} P_p$

Now, remaining animals can be arranged in any cage in ${}^{m-p} P_{m-p}$ ways

$$\therefore \text{Desired number of ways} = {}^{m-n} P_p \times {}^{m-p} P_{m-p}$$

26. (c)



$$\text{Required number of arrangements} = {}^6 P_5 \times 4! \\ = 720 \times 24 = 17280$$

27. (b) Given that, Student council has total numbers = 10

To select one President, One Vice-President, One Secretary, One Joint-Secretary and two Executive Committee members from 10 members.

Firstly, we have to select 6 members from 10 members. Which can be done in total number of ways = ${}^{10} C_6$

But there are two Executive repeated members.

$$\therefore \text{That number of ways will be} = \frac{6!}{2!}$$

Hence, total number of ways in which we can appoint

$$\text{three members} = {}^{10} C_6 \times \frac{6!}{2!}$$

$$= \frac{10!}{4!6!} \times \frac{6!}{2!} \left[\because {}^n C_r = \frac{n!}{(n-r)!r!} \right]$$

$$= \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5}{2}$$

$$= 10 \times 9 \times 8 \times 7 \times 3 \times 5 = 75600$$

28. (c) According to question, following cases arises nine digit number (less than 2×10^8) having only digits 1 and 2 = 2^8 [As first place as only one choice i.e., number 1 and rest have two choices]

Eight digit number (less than 2×10^8) having only one digit 1 and 2 = 2^8

Similarly, seven digit numbers = 2^7 .

Six digit numbers = 2^6 .

Five digit numbers = 2^5 ,

Four digit numbers = 2^4 , three digit numbers = 2^3 ,

Two digit numbers = 2^2 , One digit numbers = 2

So, numbers of all such natural numbers.

$$= 2 + 2^2 + 2^3 + 2^4 + 2^5 + 2^6 + 2^7 + 2^8 + 2^8$$

$$= 2 + 4 + 8 + 16 + 32 + 64 + 128 + 256 + 256 = 766$$

29. (d) Given word QUEEN

Starting with letters E the total number of words from = $4! = 24$

$$\text{Starting with N} = \frac{4!}{2!} = 12$$

$$\text{Starting with QE} = 3! = 6$$

$$\text{Starting with QN} = \frac{3!}{2} = 3$$

$$\text{The position of word QUEEN} = 24 + 12 + 6 + 3 + 1 = 46$$

30. (c) Continue

D				
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Positive fixed D at first place then remaining 4 letters can be arranged at 4 place = $4! = 24$

31. (b)

3 Particular never attend the parts so we can exclude those 3 from 10 friends now it is total 7 friends from which we have to invite
 $6 = {}^7C_6 = 7$

32. (b) Number of ways to distribute 10 identical prizes to 4 boys, so that each get at least one.

$$= {}^{n-1}C_{r-1} = {}^{10-1}C_{4-1} = {}^9C_3 = 84$$

33. (a) $2^{50} - 1$

34. (a) Refer to solution 4.

35. (b) The candidate is unsuccessful if he fails in 9 or 8 or 7 or 6 or 5 papers.

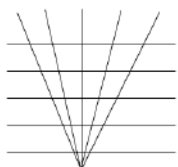
∴ The number of ways to be unsuccessful

$$= {}^9C_9 + {}^9C_8 + {}^9C_7 + {}^9C_6 + {}^9C_5$$

$$= {}^9C_0 + {}^9C_1 + {}^9C_2 + {}^9C_3 + {}^9C_4$$

$$= \frac{1}{2} ({}^9C_0 + {}^9C_1 + \dots + {}^9C_9) = \frac{1}{2} 2^9 = 256.$$

36. (c)



$$\text{Total intersection } 5 \times 5 + 1 = 26$$

37. (a) 9 bottle labelled 1,2,3....9

9 boxes labelled 1,2,39

5 bottle go their corresponding no = 9C_5

And other 4 bottle are not in their corresponding no = D4

$$= 4! \left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} \right]$$

$$= 4! \left[\frac{12-4+1}{24!} \right]$$

$$= 9$$

$$\text{Total ways} = 9 \times {}^9C_5$$

38. If married couple together total way = 7C_3

If married couple neither lengths = 7C_5

$$P(E) = \frac{{}^7C_3 + {}^7C_5}{{}^9C_5} = \frac{4}{9}$$

39. (b)

Hundreds place (can't be 0): 4

choices → {1, 2, 3, 5} Tens place: any of the 5 digits

choices 5

Units place: any of the 5 digits → 5 choices

$$\text{Total combinations} = 4 \times 5 \times 5 = 100$$

Answer: 100 (option b)

40. (b)

The word BANGLORE has 8 letters. If the substring "ANGLE" must appear together, treat it as a single block.

Letters remaining after grouping "ANGLE" are: B, O, R

So, we are arranging these 4 items: [ANGLE], B, O, R

Number of permutations = $4! = 24$.

41. (d)

Divisible by 6 ⇒ even & sum of digits divisible by 3

Digits: {1,2,3,4,5,6} → Residues mod 3:

$$\{1,4\} \rightarrow 1, \{2,5\} \rightarrow 2, \{3,6\} \rightarrow 0$$

Case last digit 2: Need one from mod 0 & one from mod 1 → $2 \times 2 \times 2 = 8$

Case last digit 4: Need one from mod 0 & one from mod 2 → $2 \times 2 \times 2 = 8$

Case last digit 6: Need one from mod 1 & one from mod 2 → $2 \times 2 \times 2 = 8$

Total = 24 → (d)